Unit 5, Lesson 3: Points on the Coordinate Plane – Part 1

Benchmark

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x- or y-axis as the line of reflection when two ordered pairs have an opposite x- and y-coordinate.

Learning Targets

- ☐ I can plot rational numbers in the form of fractional ordered pairs.
- ☐ I can plot rational numbers in the form of decimals.

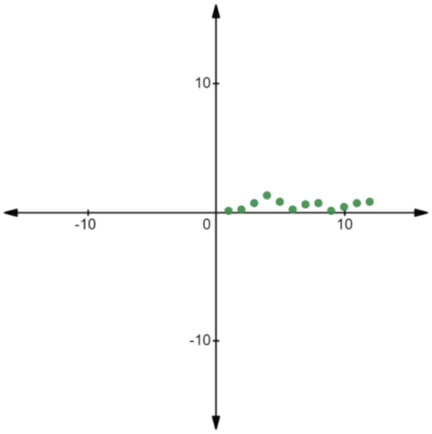
5.3.1 Warm-Up: Which One is the Best?

Coral polyps get stressed out and could die if the surrounding water is above 29℃. The table represents the average temperature, to the nearest tenth, above 29℃ at the Great Coral Reef over a 12-week period.

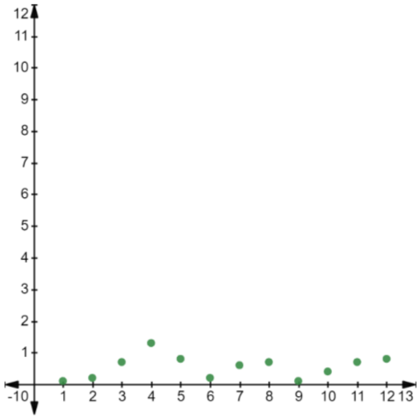
1. Explain which of the three graphs best represents the data. (Data taken from NOAA.)

Week	1	2	3	4	5	6	7	8	9	10	11	12
Temp	0.1	0.2	0.7	1.3	0.8	0.2	0.6	0.7	0.1	0.4	0.7	0.8

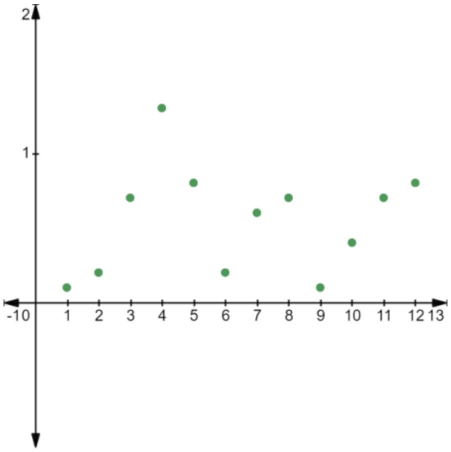
Graph A



Graph B

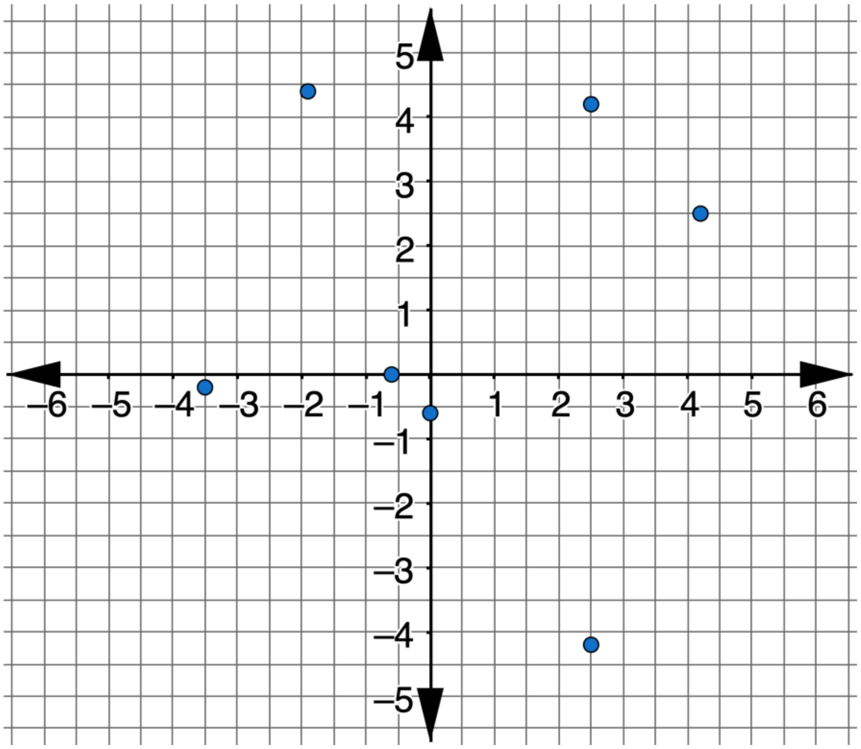


Graph C



5.3.2 Collaboration: Plotting Points with a Decimal Scale

1. A coordinate plane is shown.



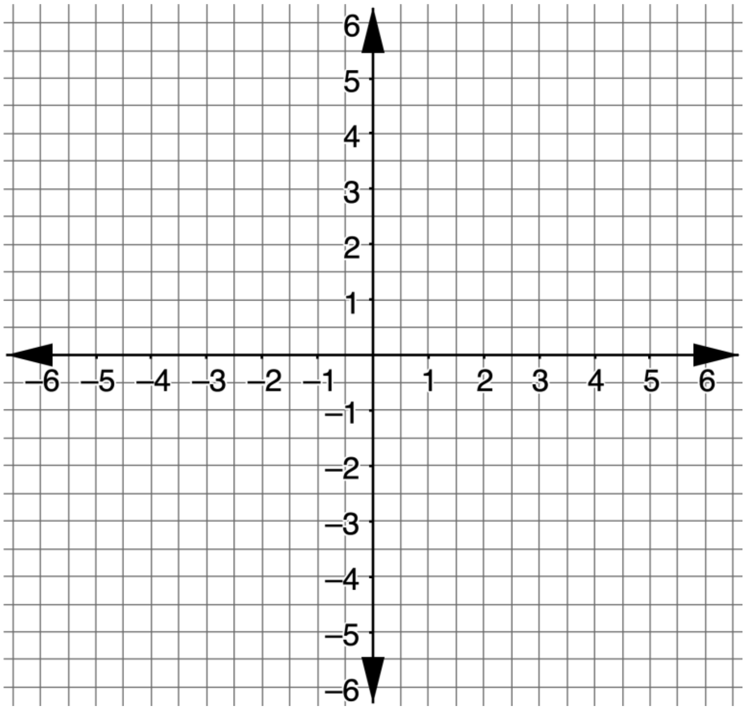
a. Working with your partner, describe the scale that is being used for each axis.

b. Locate each point on the coordinate plane and label it with the appropriate letter.

Point G $(4.2, 2.5)$	Point B $(-1.9, 4.4)$	Point F $(0, -0.6)$	Point W $(-3.5, -0.2)$
Point E $(-0.6, 0)$	Point A $(2.5, 4.2)$	Point M $(2.5, -4.2)$	

2. Working with your partner, plot and label the points using the given letters on the grid.

Point L $(1.5, 2)$	Point B $(3.9, -4.5)$	Point G $(-5.5, -2.1)$	Point D $(0, 0.7)$
Point K $(-2.8, 5.7)$	Point S $(-0.8, 0)$	Point N $(-1, -3.4)$	Point H $(-3.9, 4.5)$



a. Which points were easy to plot?

b. Which points were more difficult to plot?

3. On your own, answer the following question:

When plotting ordered pairs with decimal values, how is it best to determine the scale to use when setting up the axes on a grid?

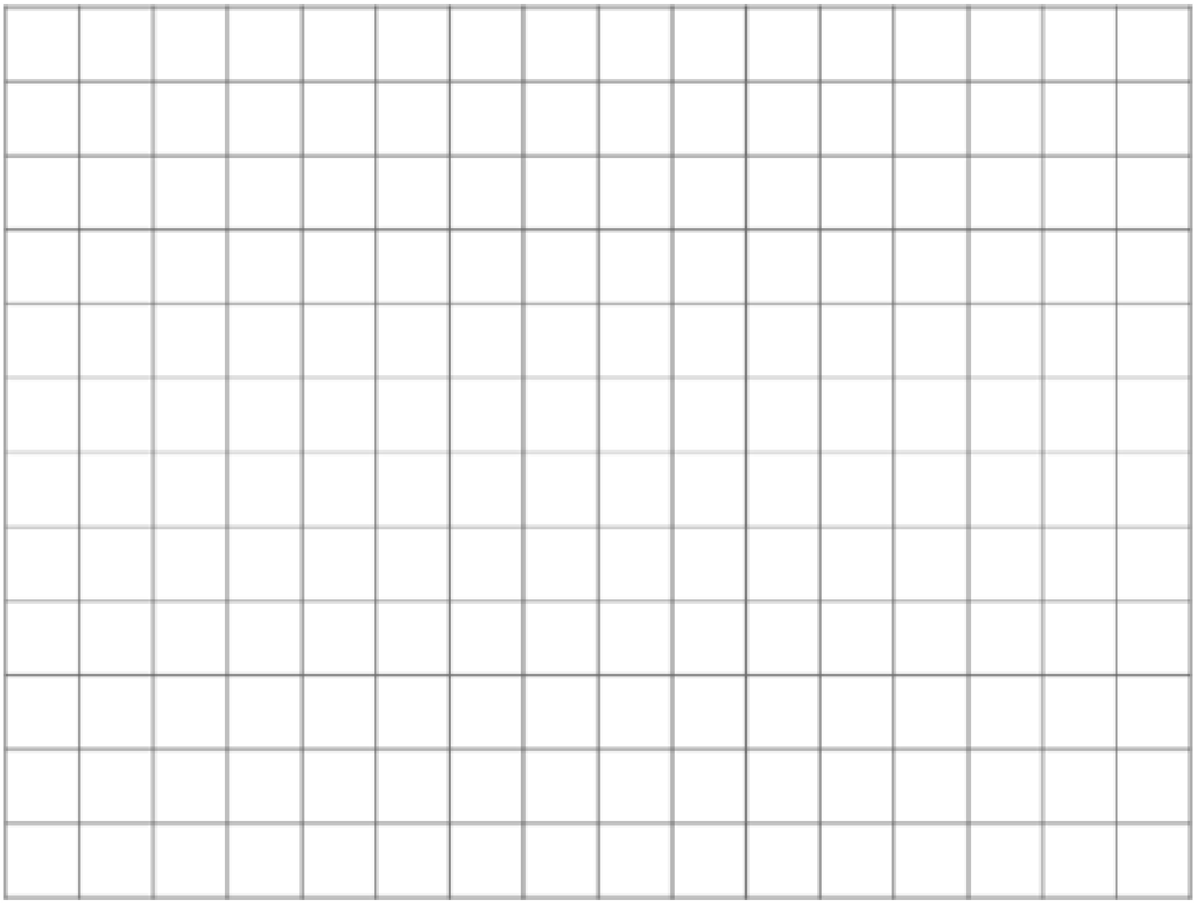
My Reasoning

Share your reasoning with your partner and ask for feedback on any parts that were unclear or missing. Make any necessary edits to your reasoning.

5.3.3 Try It: Plotting Points with a Decimal Scale

1. When plotting $(1.75, -2.5)$ and $(-2.25, 1.5)$, describe how to best scale the axes.

2. For the points $(1.75, -2.5)$ and $(-2.25, 1.5)$, draw and label coordinate axes with an appropriate scale, and plot the points.



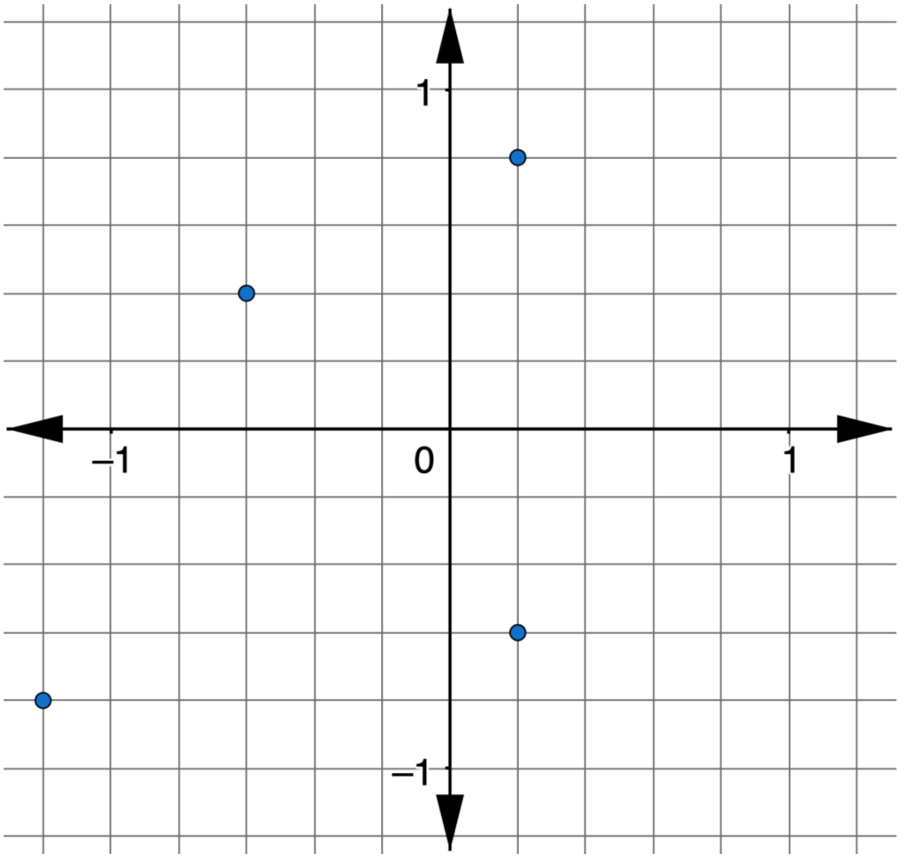
5.3.4 Guided Instruction: Plotting Points with a Fractional Scale

1. Four points are shown.

Point L	Point H	Point S	Point R
$\left(\frac{1}{5}, \frac{4}{5}\right)$	$\left(-\frac{3}{5}, \frac{2}{5}\right)$	$\left(-1\frac{1}{5}, -\frac{4}{5}\right)$	$\left(\frac{1}{5}, -\frac{3}{5}\right)$

a. Based on the denominators of each fraction, what scale should be used for the grid?

b. Find and label the points with the given letters on the coordinate plane.



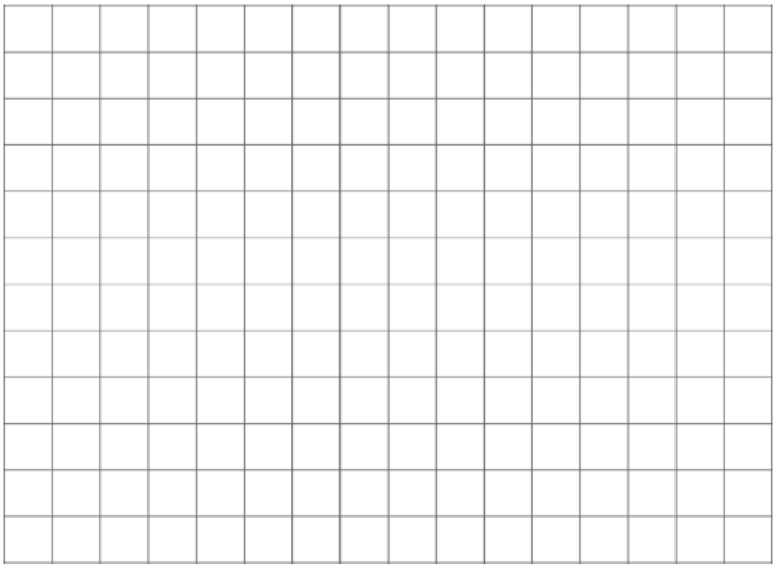
2. Four different points are shown.

Point F	Point T	Point S	Point D
$\left(\frac{1}{2}, \frac{-3}{4}\right)$	$\left(1\frac{1}{4}, \frac{1}{2}\right)$	$\left(-\frac{1}{4}, -\frac{1}{2}\right)$	$\left(-1\frac{3}{4}, 1\right)$

a. List the denominators of each fraction.

b. Liana says when she sets up her grid, she will use a scale of $\frac{1}{4}$. Trey says he plans to use a scale of $\frac{1}{2}$. Explain who is planning to use the better scale.

c. Draw and label coordinate axes with the scale of your choice. Then, plot and label the points.



3. Another set of four points is shown.

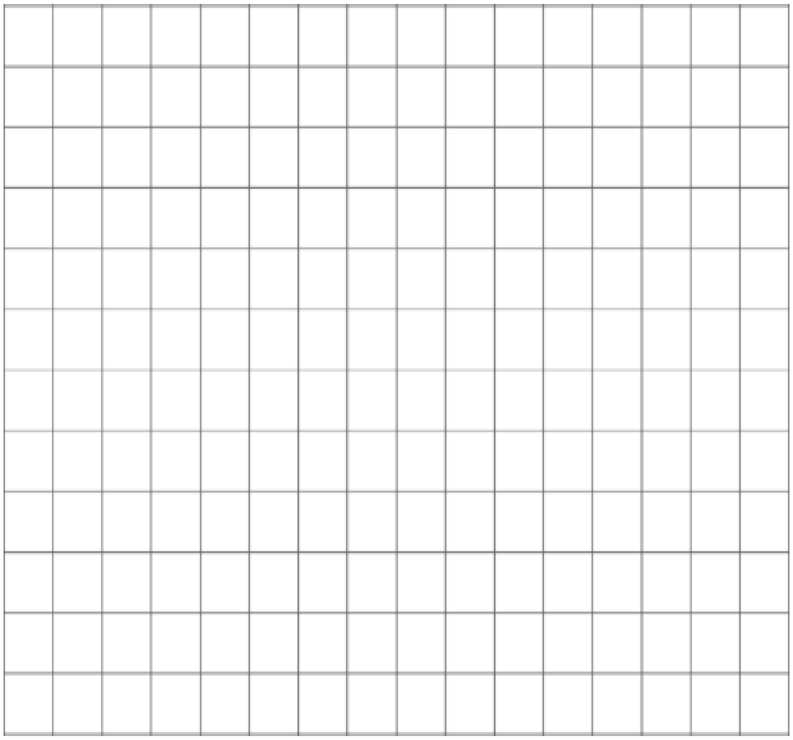
Point M	Point L	Point C	Point A
$\left(\frac{2}{3}, \frac{1}{6}\right)$	$\left(\frac{1}{2}, -\frac{1}{3}\right)$	$\left(-\frac{5}{6}, -\frac{1}{2}\right)$	$\left(-\frac{1}{6}, 1\right)$

a. List the denominators of each fraction.

b. Zoe says she does not want to label her axes by halves because she will only be able to approximate the location of the thirds. She also says she does not want to label by thirds because there is no easy way to approximate the location of the halves. Christopher says she should label her axes by sixths.

Explain the benefit of using Christopher’s plan.

c. Draw and label coordinate axes with the scale you chose and plot the points.

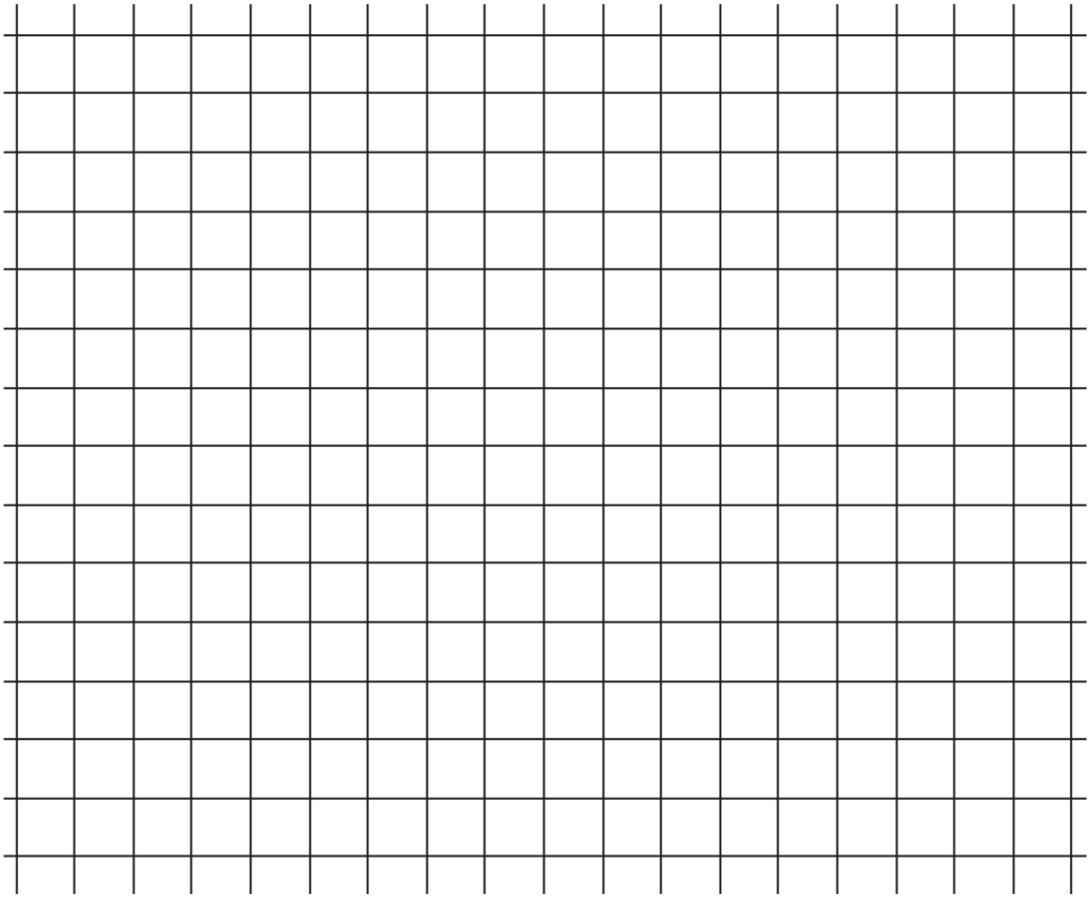


When deciding on a scale for plotting ordered pairs with rational values, consider the least common multiple (LCM) of each of the denominators to determine the proper increments.

5.3.5 Try It: Plotting Points with Fractions as a Scale

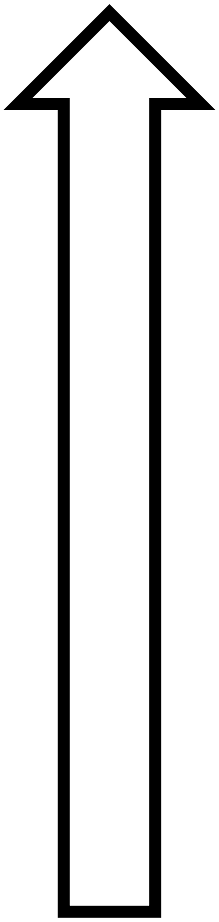
1. Draw and label coordinate axes with an appropriate scale. Then, plot and label the points.

Point W $\left(\frac{2}{3}, \frac{1}{4}\right)$	Point D $\left(\frac{1}{2}, -\frac{1}{3}\right)$	Point P $\left(-\frac{5}{3}, -\frac{1}{2}\right)$	Point T $\left(-\frac{1}{6}, \frac{7}{4}\right)$



5.3.6 Wrap-Up: Reflection

1. Shade the arrow up to the statement that best describes your current level of understanding.



I'm so confident, I could explain plotting ordered pairs made up of fractions and/or decimals to someone else!

I can get to the right answer, but I don't understand it well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked the boxes I think I do not understand.

I tried hard and I listened, but I am finding this challenging. I checked the boxes I think I do not understand.

2. Check the skills that you currently understand.

- ☐ Determining the correct scale
- ☐ Identifying the LCM of the denominators for coordinates that include fractions
- ☐ Plotting ordered pairs that contain fractions
- ☐ Plotting ordered pairs that contain decimals